Grape Segmentation

(Comparative Analysis of YOLO Models for Grape Segmentation)

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Abstract:

Grape segmentation is a key feature of precision farming that helps with things like figuring out how much fruit to pick and how much to expect from a crop. But in vineyards, occlusion, fluctuating light, and busy backgrounds make it challenging to get exact segmentation.
For a consistent evaluation, all models were trained on identical dataset partitions, image resolutions, and hyperparameter settings.

Performance was assessed using mask-level precision, recall, F1-score, mAP@0.5, mAP@0.5:0.95, and inference latency metrics. The test outcomes indicate that YOLOv11 exhibited superior mAP and precision values, thereby demonstrating the highest segmentation accuracy. In contrast, YOLOv26 showed a favorable balance between accuracy and speed. It achieved the fastest inference speed while still performing well.
In contrast, YOLOv12's efficacy was diminished due to the absence of pre-trained segmentation weights.

These results show that model selection should strike a balance between speed and accuracy, based on the needs of real-world agricultural systems.

Keywords: Grape Segmentation, YOLO, Deep Learning, Precision Agriculture, Instance Segmentation

1 Introduction

Precision agriculture has become a crucial area for enhancing productivity, resource utilization, and environmental sustainability in contemporary agricultural practices. Accurate fruit segmentation is essential for applications including yield estimation, disease identification, and automated harvesting within this framework. Specifically, in vineyard settings, dependable segmentation of grape clusters is of paramount importance, given its direct influence on subsequent decision-making processes and automation systems.

On the other hand, segmenting grapes in real-world conditions is quite challenging. Images from vineyards often have complex backgrounds, changing lighting, and objects like leaves or overlapping grape clusters. Because of these factors, it becomes difficult to get consistent and reliable results using traditional methods.

Thresholding, edge detection, and colour-based methods worked well in controlled environments. However, in real-world conditions where lighting and backgrounds keep changing, their performance decreases.Consequently, researchers are increasingly embracing deep learning methods. These approaches have the ability to autonomously extract valuable features from the data.YOLO (You Only Look Once) models have gained considerable traction due to their speed and efficiency. Subsequent iterations of YOLO have demonstrated enhancements in both precision and processing velocity, thereby rendering them appropriate for application in authentic agricultural settings, including on devices with constrained computational capabilities.Even with these improvements, there is still no proper comparison of the latest YOLO models for grape segmentation under the same conditions.

This study aims to fill this gap by comparing three recent YOLO-based segmentation models—YOLOv11, YOLOv12, and YOLOv26—using the same experimental setup. The models are evaluated for both accuracy and speed to assess their performance. The main goal is to help choose the best model for practical use in agriculture.

2 Methodology

A labelled vineyard image dataset was employed for training and validation under controlled experimental conditions. Each model was trained using consistent hyperparameters to minimise experimental bias. Performance was assessed through precision, recall, F1-score, and mean Average Precision metrics, alongside inference time measurements. In doing so, the analysis considers not only detection capability but also practical deployability in real-world agricultural systems.

2.1 Dataset

The proposed models were trained and evaluated using a labeled grape dataset. This dataset comprises vineyard images acquired in authentic environmental settings, encompassing fluctuations in illumination, background intricacy, and obstructions stemming from foliage.

Instance segmentation annotations were applied to all images, categorizing them into three distinct classes. To facilitate a just and impartial comparison, the dataset was partitioned into training, validation, and test sets, employing uniform splits across all models. The identical dataset configuration was consistently utilized throughout all experimental procedures.

2.2 Models

This study evaluates three modern YOLO-based instance segmentation models: YOLOv11, YOLOv12, and YOLOv26.These models are part of the YOLO family, which is recognized for its capacity to perform object detection and segmentation in real-time.YOLOv11 and YOLOv26 were initialized with pretrained segmentation weights, facilitating quicker convergence and enhanced feature extraction. Conversely, YOLOv12 was trained from itsarchitectural configuration, given the unavailability of publicly accessible pretrained segmentation weights. This disparity in initialization is taken into account during the performance analysis of each model.

2.3 Training Setup

To keep things consistent, all models were trained using the same settings. The training ran for 500 epochs with image size set to 640×640 and a batch size of 16. Standard Ultralytics settings were used for optimization and data augmentation. All training was carried out on a GPU.

2.4 Evaluation Metrics

Performance was evaluated using precision, recall, F1-score, mAP@0.5, and mAP@0.5:0.95. In addition, inference time per image was measured to assess computational efficiency.

3.1 Quantitative Results

Model	Precision	Recall	F1-score	mAP@0.5	mAP@0.5:0.95	Inference Time (ms)
YOLOv11	**0.599**	**0.481**	**0.534**	**0.525**	**0.284**	**21.05**
YOLOv12	**0.230**	**0.399**	**0.292**	**0.303**	**0.137**	**20.62**
YOLOv26	**0.478**	**0.513**	**0.495**	**0.411**	**0.227**	**16.81**

Table 1: Performance comparison of YOLO models for grape instance segmentation using mask-level evaluation metrics.

3.2 Performance Analysis

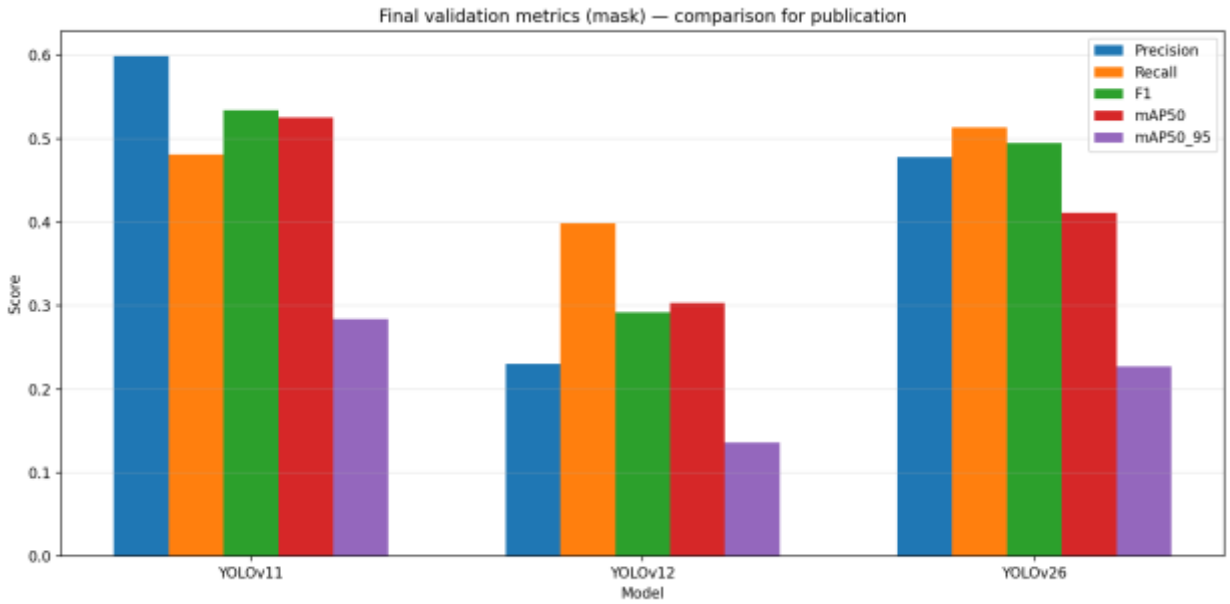
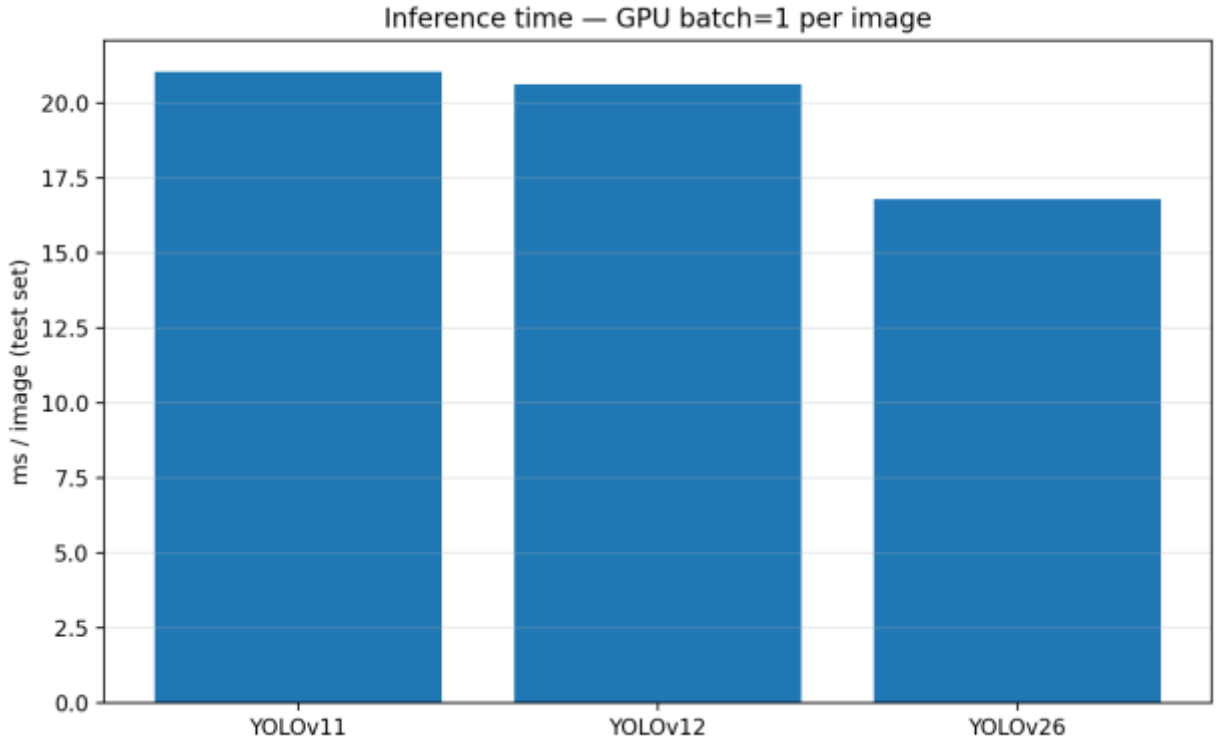


Figure 1: Qualitative comparison of grape instance segmentation results using YOLOv11, YOLOv12, and YOLOv26 under complex vineyard conditions.

3.3 Inference Speed



3.4 Quantitative Results

Table 1 presents the quantitative performance metrics for the models under evaluation. YOLOv11 exhibits the most favorable segmentation accuracy, attaining the highest mAP@0.5 (0.525) and mAP@0.5:0.95 (0.284) scores. Furthermore, it achieves the highest precision (0.599), suggesting more dependable predictions with a reduced incidence of false positives. Conversely, YOLOv26 displays a more equilibrated performance profile, achieving the highest recall (0.513), which signifies improved detection of grape instances. Moreover, it offers the most rapid inference speed (16.81 ms per image), thereby rendering it appropriate for real-time applications.In contrast, YOLOv12 demonstrates diminished performance across the majority of evaluation metrics, including precision (0.230) and mAP values. This is likely attributable to the absence of pretrained segmentation weights during the training phase.

Figure 2: Qualitative comparison of grape instance segmentation results using YOLOv11, YOLOv12, and YOLOv26 under complex vineyard conditions.

4 Discussion

4.1 Looking at how well grape segmentation works

The results show that the YOLO models work in different ways, even if they are trained the same. It is hard to see grapes in vineyards because the background is messy, the light keeps changing, and grapes are covered by leaves. Because of this, the models find it hard to detect and separate the grapes correctly.Out of all the models we've tried, YOLOv11 is the best. It is the most accurate based on mAP scores. This means that it can generate segmentation masks that are more precise and find grapes more easily. One reason it works well is because it uses pretrained weights, which help the model learn better features from the start.On the other side, YOLOv26 is better overall. It has the best recall, however YOLOv11 is a little more accurate. This means it can find more grape clusters, even the ones that are only partly hidden. This is very important in real vineyards because if grapes are missed, it can affect yield estimation and harvesting time.YOLOv12 does the worst job of all of them. It is less accurate and has problems telling the grapes from the backdrop. This is largely because it doesn't use pretrained segmentation weights, which implies it can't learn strong features very effectively.

4.2 The balance between speed and accuracy

One important thing to remember about the findings is that there is a balance between speed and precision.YOLOv11 is the most accurate, but it takes a little longer to make inferences.However, YOLOv26 is faster and has the smallest inference time, which makes it suitable for application in real time.This means that the model you pick will depend on how you want to use it.If you need very exact results, as when you want to know if someone has a condition or do a full analysis, YOLOv11 is the superior choice.YOLOv26 is preferable if speed is more important, as when you need to monitor something in real time or pick crops with a robot.

4.3 A qualitative analysis of segmentation outcomes

Figure 2 indicates that YOLOv11 generates segmentation masks that are cleaner and more accurate, with fewer errors. This is especially true for grape clusters that are close together.YOLOv26 is better at finding grapes that are partly hidden, and it works well even when things are tricky.YOLOv12 is not as good as the others. It misses many grapes and the shapes it makes are not very accurate.These results show that it is very important for the model to learn good features and start properly when working with real-world farm images.

4.4 Issues and Future Actions

This study has some issues.

All the experiments used the same dataset and training setup, so the results might not work the same for different grapes or vineyard conditions.

Also, YOLOv12 did not use pretrained weights, which makes the comparison slightly unfair.

In the future, this will work better if you use bigger and more diversified datasets. Adding data in better ways can also make things work better.

Researchers can also try using lightweight models for devices like drones or robots in farming, or combine multiple models to get better performance.

5 Conclusion

Just making changes in the model structure does not mean it will perform better everywhere. For agricultural segmentation work, we must check both how accurate the model is and how practical it is to use, such as its speed and system requirements. This study also explains how common object detection models can be applied to agriculture-based tasks.

6 Refrences

1. Kamilaris, A., & Prenafeta-Boldú, F. (2018). *Deep learning in agriculture: A survey*. Computers and Electronics in Agriculture.
2. Sa, I. et al. (2016). *DeepFruits: A fruit detection system using deep neural networks*. Sensors.
3. Bargoti, S., & Underwood, J. (2017). *Deep fruit detection in orchards*. ICRA.
4. Joseph Redmon et al. (2016). You Only Look Once: Unified, Real-Time Object Detection. CVPR.
5. Kaiming He et al. (2017). Mask R-CNN. ICCV.
6. Redmon, J.; Divvala, S.; Girshick, R.; Farhadi, A. You only look once: Unified, real-time object detection. In *Proceed
7. Liu, S.; Whitty, M. Automatic grape bunch detection in vineyards with an SVM classifier. J. Appl. Log. 2015, 13, 643–653.
8. Behroozi-Khazaei, N.; Maleki, M.R. A robust algorithm based on color features for grape cluster segmentation. Comput. Electron. Agric. 2017, 142, 41–49.
9. Mohimont, L.; Roeseler, M.; Rondeau, M.; Gaveau, N.; Ain, F.; Steffenel, L.A. Comparison of machine learning and deep learning methods for grape cluster segmentation. In Proceedings of the International Conference on Smart and Sustainable Agriculture, Virtual Event, 21–22 June 2021; pp. 84–102.